Fiscal Year:	FY 2021	**Task Last Updated:**	FY 04/05/2021
PI Name:	Ganguli, Gurudas Ph.D.		
Project Title:	Understanding the Frequency Synchronization Physics in PK-4 Experiment		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Washington	**State:**	DC
Zip Code:	20375-0001	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2016 Physical Sciences NNH16ZTT002N: Fundamental Physics – NASA/NSF Partnership on Science of Dusty Plasmas: Utilizing the PK-4 Facility on board the ISS
Start Date:	04/15/2017	**End Date:**	12/31/2023
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
Contact Monitor:	Callas, John	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2023 per U. Israelsson/JPL (Ed., 1/6/22) NOTE: End date changed to 10/15/2021 per U. Israelsson/JPL (Ed., 4/6/21)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Crabtree, Chris Ph.D. (Naval Research Laboratory)		
Grant/Contract No.:	JPL 1573108		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The objective of this proposal is to provide theoretical support for a separate PK-4 (Plasma Krystall-4) experimental project, “Nonlinear Waves in Dusty Plasmas.” The experimental effort, led by Principal Investigator (PI) Goree, is an investigation of nonlinear synchronization, with the advance that it will study not point oscillators, but waves that propagate a distance of many wavelengths. The literature is lacking theory suitable for comparison to the experiment, perhaps because the experiments at this point are still very new. In particular, theory is needed for the case, beyond the point oscillator description that is most common, for synchronization in the realm of propagating waves. Beyond just supporting the PK-4 experiment, the theoretical advances we propose can have a broad interdisciplinary impact. Since synchronization, frequency entrainment, and frequency pulling are nonlinear phenomena that impact a variety of physical, chemical, and biological systems, the proposed investigation will be relevant to a broader area of contemporary interest and enhance the science foundation for nonlinear phenomena in plasma physics. A plasma system normally constitutes a highly complex medium that can display a rich variety of nonlinear phenomena arising out of a host of intrinsic oscillations and other non-stationary states. A clear identification of phenomena like synchronization or frequency entrainment in such a medium often poses serious experimental challenges. Many past studies on such systems have therefore relied on driving a single plasma system with an external frequency source. There have been very few studies devoted to the investigation of mutual synchronization between two plasma systems except where this phenomenon was studied between two double plasma devices and where it was investigated in two electrical discharges. There is no theoretical framework to account for these phenomena in a realistic dispersive plasma system that includes natural growth and dissipation of waves. Therefore, the primary focus of this proposal is to develop such a framework and apply it to understand the dedicated PK-4 dusty plasma experiment on frequency synchronization. The validated model will then be useful for general application to other experiments. Some of the science questions that we will address in our proposed theoretical investigation are the following: • What are the parametric domains in the state space of the external driver frequency and amplitude that lead to frequency entrainment, frequency pulling, and synchronization of the dust acoustic waves in the PK-4 plasma? • How do dispersion and nonlinearity of the dusty plasma medium influence the nature of the synchronization process? • Can the non-harmonic synchronized state, as observed in ground-based experiments, be predicted by a continuum model of the dusty plasma's constituents? • What determines the threshold amplitude for synchronization, and how is the threshold affected by dust-dust correlations (i.e., the strength of the Coulomb parameter)? • How does viscous dissipation in the medium affect synchronization and influence the nature of the synchronized excitation? • Can one observe phase-flip transitions in the synchronization, i.e., a sudden transition between in-phase synchronization to pi out-of-phase synchronization? • How does a broad-spectrum driver (e.g., signals from another plasma source) alter the synchronization scenario?
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Beyond just supporting the PK-4 experiment, the theoretical advances in this project will have a broad interdisciplinary impact. Since synchronization, frequency entrainment, and frequency pulling are nonlinear phenomena that impact a variety of physical, chemical, and biological systems, the investigation is relevant to a broader area of contemporary interest.
Task Progress:	Progress was made in two areas of interest towards understanding the salient physics of the PK-4 experiments. 1. Synchronization Model Studies: Following the insight gained from our numerical investigations over the past year – particularly regarding the importance of the existence of a limit cycle in the model equation – we have focused our efforts towards the exploration of synchronization phenomenon in a KdV-Burger type model. Such a model is appropriate for studying nonlinear propagation in a dusty plasma when dissipative losses arising from viscosity effects are important. We are studying the driven response of such a model for a variety of time and space dependent sources such as sinusoidal drivers, conoidal wave drivers, and sawteeth shaped drivers. A parametric plot of the synchronization regimes in the form of an Arnold tongue diagram is being constructed for various equilibrium characteristics of the dusty plasma. This will aid us in designing and guiding such experiments to be carried out on the PK-4 device. A manuscript on this work is currently under preparation. 2. Nonlinear Mixing: As mentioned last year, we had also commenced studies on examining the driven response of a KdV equation (in the absence of dissipation). This work has been completed. We have found that for a time varying sinusoidal driving there exists an exact analytic solution of the model which shows the existence of nonlinear wave mixing in a dusty plasma medium. Physically, the model represents wave mixing arising from the temporal modulation of a nonlinear dust compressional wave. This is a situation that can be easily realized in an experiment using radiation pressure of lasers or time varying electric potentials to modulate self-excited or externally driven large amplitude compressional waves. The solution not only shows the existence of the wave mixing phenomenon in this simple model system but may also be useful in predicting nonlinear wave mixing for a proposed experimental configuration in a two-dimensional dusty plasma medium. Our work has been published in the Physics of Plasmas [Phys. Plasmas 27, (2020) 113701)] and was announced to be the Editor’s Pick [Ed. note: see also Bibliography section]. Papers published in refereed journals: Mir A.A., S.K. Tiwari, J. Goree, A. Sen, C. Crabtree, and G. Ganguli, “A forced Korteweg-de Vries model for nonlinear mixing of oscillations in a dusty plasma”, Phys. Plasmas 27, 113701 (2020); https:// Manuscripts Under Preparation:

	Mir A.A., S.K. Tiwari, J. Goree, A. Sen, C. Crabtree, and G. Ganguli “A driven Korteweg-de Vries-Burger model for synchronization of dust acoustic waves” Invited talks at conferences: Invited talk on “Nonlinear mixing of oscillations in dusty plasma” (delivered by S.K. Tiwari) at the International e-Conference on Plasma Theory and Simulations (PTS-2020), 14 & 15 September 2020, Bilaspur (India)
Bibliography Type:	Description: (Last Updated: 03/22/2023)
Articles in Peer-reviewed Journals	Mir AA, Tiwari SK, Goree J, Sen A, Crabtree C, Ganguli G. "A forced Korteweg-de Vries model for nonlinear mixing of oscillations in a dusty plasma." Physics of Plasmas. 2020 Nov;27(11):113701. https://doi.org/10.1063/5.0022482 , Nov-2020
Awards	Mir AA, Tiwari SK, Goree J, Sen A, Crabtree C, Ganguli G. "Editor's Pick November 1, 2020 for article 'A forced Korteweg-de Vries model for nonlinear mixing of oscillations in a dusty plasma.' Physics of Plasmas. 2020 Nov;27(11):113701. https://doi.org/10.1063/5.0022482 " Nov-2020